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Roll No : 150096724023

Subject : Devops

Problem Statement No : 118

Case Study 118: Project NeuroCompute – Distributed AI Supercomputing Platform

Github Repo Link :
https://github.com/RaginiSingh2024/NeuroCompute-DevOps

Documentation link :
https://docs.google.com/document/d/1tR_-45rCkCp58_c0MmpDdHVJc_DYIuPgB09cC_EmoDs/edit?usp=sharing

View Demo :

NeuroCompute DevOps Platform

Cloud Computing & DevOps Project Report

Abstract

Project NeuroCompute – Distributed AI Supercomputing Platform is a cloud-native platform designed to simulate the deployment and management of distributed AI workloads using modern DevOps and Cloud Computing practices. The project integrates React, Node.js, MongoDB, Docker, Jenkins, Kubernetes, Terraform, Prometheus, Grafana, ELK Stack, and HashiCorp Vault to demonstrate an end-to-end production-ready DevOps ecosystem.

Objectives

The major objectives of the project are:

- Build a modern full-stack application.
- Implement Docker containerization.
- Configure Continuous Integration and Continuous Deployment (CI/CD).
- Deploy workloads using Kubernetes.

- Automate infrastructure using Terraform.
- Implement monitoring and alerting.
- Enable centralized logging.
- Secure secrets using HashiCorp Vault.
- Follow cloud-native development practices.

Problem Statement

Modern software applications require automation, scalability, monitoring, security, and rapid deployment capabilities.

Traditional deployment approaches often suffer from:

- Manual deployment errors
- Lack of scalability
- Poor monitoring

- Difficult infrastructure management

- Weak security practices

This project addresses these challenges by implementing a complete DevOps ecosystem around a full-stack application.

System Overview

The platform consists of:

Frontend

- React.js
- Vite
- React Router
- Axios
- Framer Motion

Backend

- Node.js
- Express.js
- JWT Authentication

Database

- MongoDB Atlas

DevOps Components

- Docker
- Jenkins

- Kubernetes
- Terraform
- Prometheus
- Grafana
- ELK Stack
- HashiCorp Vault

System Architecture

The overall architecture follows a cloud-native approach.

Frontend Layer

↓

Backend API Layer

↓

MongoDB Atlas Database

↓

Docker Containers

↓

Jenkins CI/CD Pipeline

↓

Kubernetes Cluster

↓

Monitoring & Logging Stack

↓

Vault Secret Management

Technology Stack

Category	Technologies
Frontend	React, Vite, Axios
Backend	Node.js, Express.js
Database	MongoDB Atlas
Containerization	Docker
CI/CD	Jenkins
Orchestration	Kubernetes
Infrastructure	Terraform
Monitoring	Prometheus, Grafana
Logging	ELK Stack
Security	HashiCorp Vault

Version Control	Git, GitHub

Features

User Management

- User Registration
- User Login
- JWT Authentication
- Secure Password Storage
- Role-Based Access Control

Job Portal

- Browse Jobs
- Search Jobs
- Apply for Jobs
- Application Tracking

Dashboard

- User Dashboard
- Analytics View
- Monitoring Overview
- Activity Tracking

Responsive UI

- Mobile Friendly
- Modern Interface
- Fast Navigation
- Responsive Layout

Docker Implementation

Docker was used to containerize both frontend and backend applications.

Benefits

- Consistent environments
- Easy deployment
- Portability
- Isolation of services

Files

- Dockerfile.frontend
- Dockerfile.backend
- docker-compose.yml

Jenkins CI/CD Pipeline

Jenkins automates the software delivery process.

Pipeline Stages

1. Source Code Checkout
2. Dependency Installation
3. Build Process
4. Testing
5. Docker Build
6. Deployment

Benefits

- Faster deployments
- Reduced manual work
- Improved reliability

Kubernetes Deployment

Kubernetes manages application deployment and scalability.

Kubernetes Resources

- Namespace
- Deployment
- Service
- Ingress
- ConfigMap
- Secret

Advantages

- Auto-healing
- Scaling
- High availability

- Load balancing

Terraform Infrastructure

Terraform is used for Infrastructure as Code (IaC).

Files

- provider.tf
- main.tf
- variables.tf
- outputs.tf

Benefits

- Infrastructure automation
- Version-controlled infrastructure
- Repeatable deployments

Monitoring with Prometheus & Grafana

Monitoring is implemented using Prometheus and Grafana.

Prometheus

Responsibilities:

- Metrics Collection
- Alert Management
- Performance Monitoring

Grafana

Responsibilities:

- Dashboards
- Visualization
- Analytics
- Reporting

Centralized Logging with ELK Stack

The ELK Stack provides centralized logging.

Elasticsearch

- Log Storage
- Data Indexing

Logstash

- Log Processing
- Data Transformation

Kibana

- Log Visualization
- Search and Analysis

Secret Management using Vault

HashiCorp Vault is used to securely store secrets.

Managed Secrets

- API Keys
- Database Credentials
- JWT Secrets
- Configuration Variables

Benefits

- Secure storage
- Controlled access
- Secret rotation support

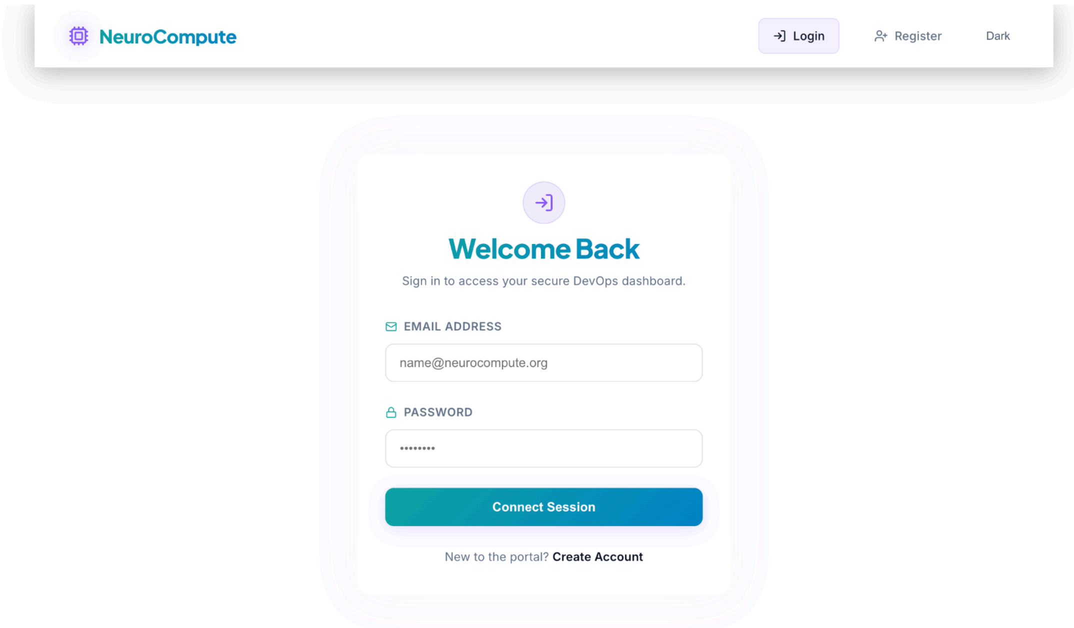
Project Structure

```
NeuroCompute_Devops
│
├── backend
├── frontend
├── docker
├── jenkins
├── kubernetes
├── terraform
├── monitoring
├── elk
├── vault
├── diagrams
└── screenshots
```

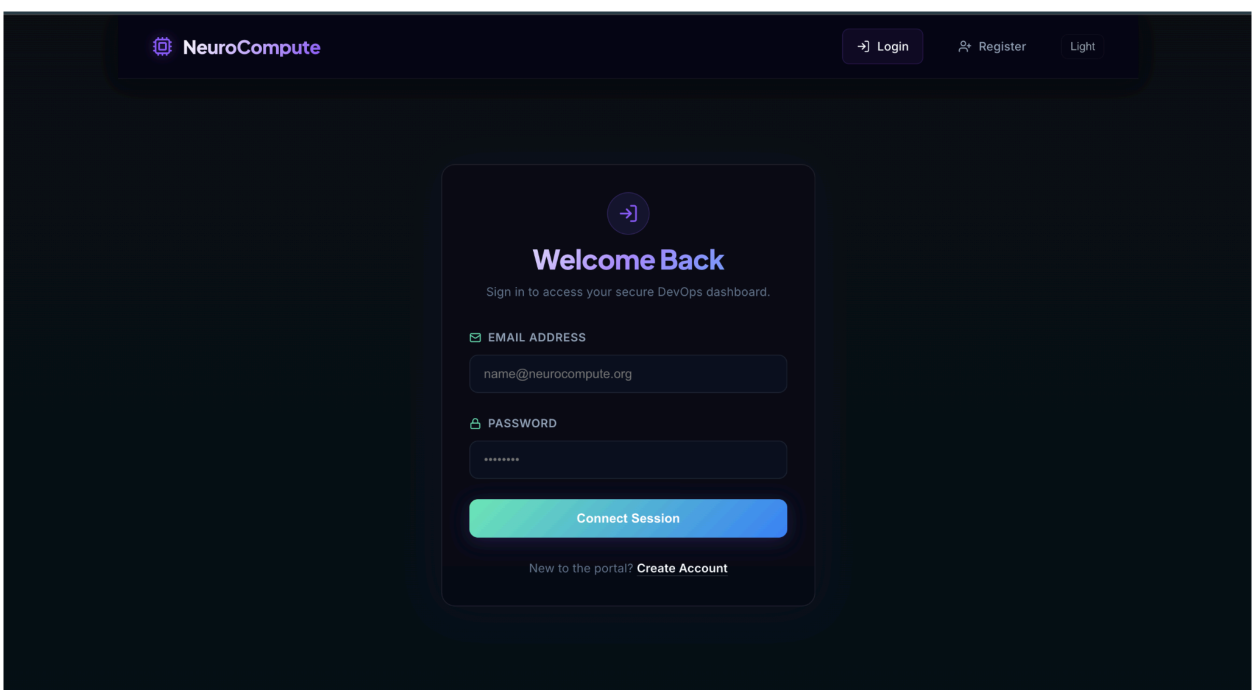
Screenshots

Insert screenshots of:

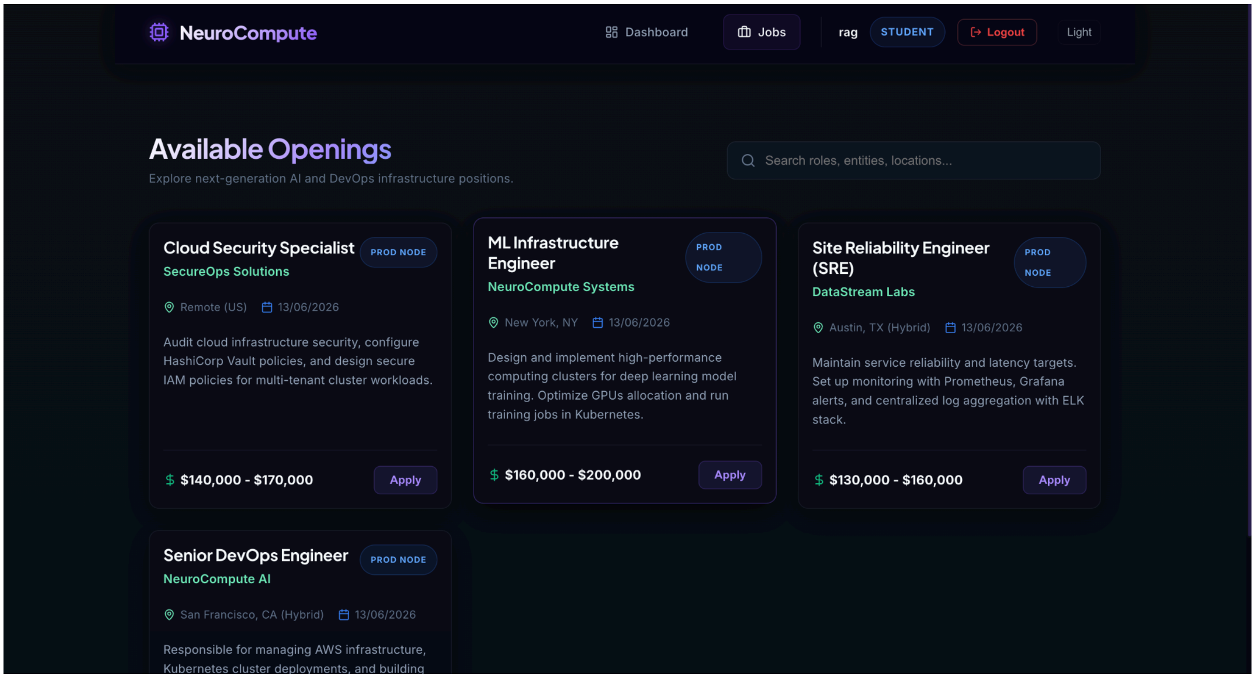
1. **Login Page**



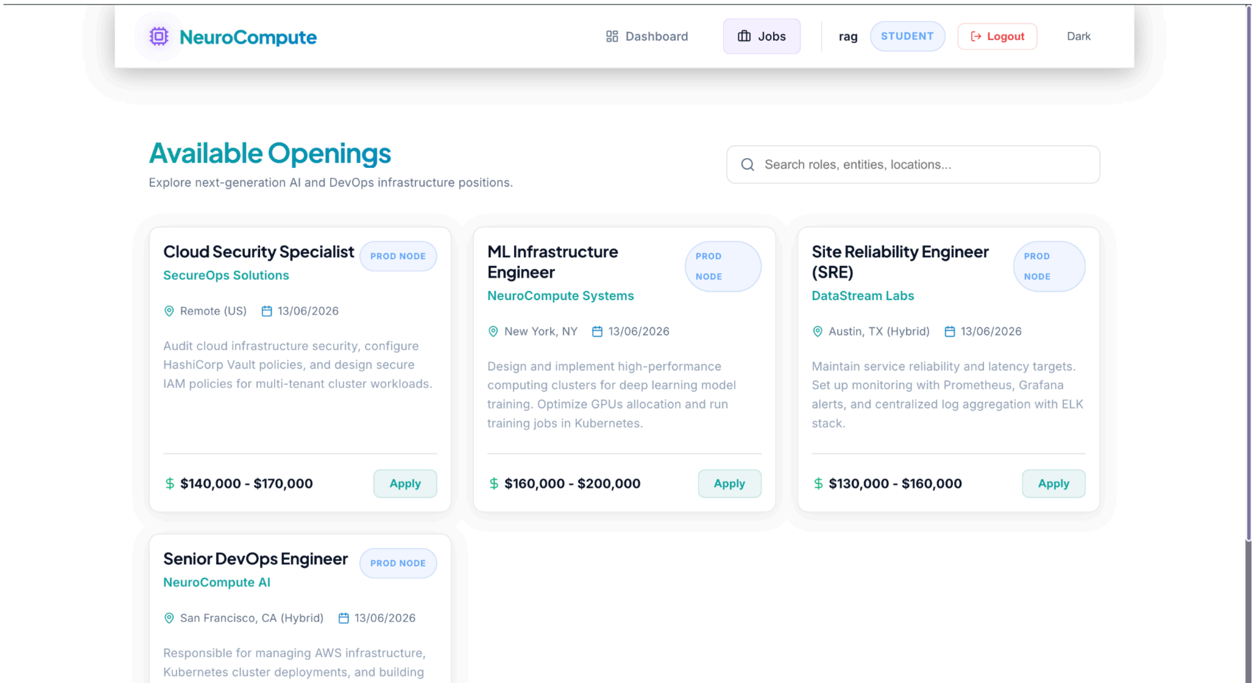
2, Login /Sign In page



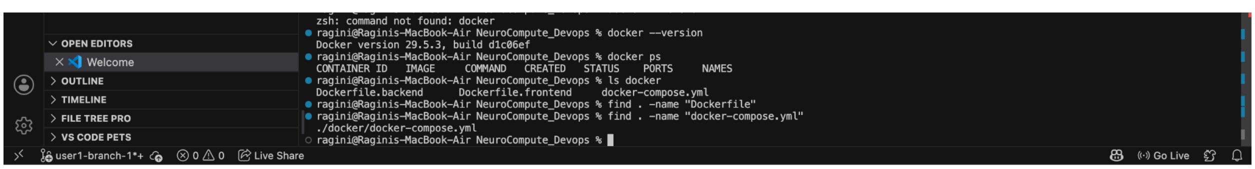
3. Job Portal


NeuroCompute
Dashboard
Jobs
rag
STUDENT
Logout
Light
Available Openings
Explore next-generation AI and DevOps infrastructure positions.
Search roles, entities, locations...
Cloud Security Specialist
PROD NODE
SecureOps Solutions
Remote (US)
13/06/2026
Audit cloud infrastructure security, configure HashiCorp Vault policies, and design secure IAM policies for multi-tenant cluster workloads.
$140,000 - $170,000
Apply
ML Infrastructure Engineer
PROD NODE
NeuroCompute Systems
New York, NY
13/06/2026
Design and implement high-performance computing clusters for deep learning model training. Optimize GPUs allocation and run training jobs in Kubernetes.
$160,000 - $200,000
Apply
Site Reliability Engineer (SRE)
PROD NODE
DataStream Labs
Austin, TX (Hybrid)
13/06/2026
Maintain service reliability and latency targets. Set up monitoring with Prometheus, Grafana alerts, and centralized log aggregation with ELK stack.
$130,000 - $160,000
Apply
Senior DevOps Engineer
PROD NODE
NeuroCompute AI
San Francisco, CA (Hybrid)
13/06/2026
Responsible for managing AWS infrastructure, Kubernetes cluster deployments, and building

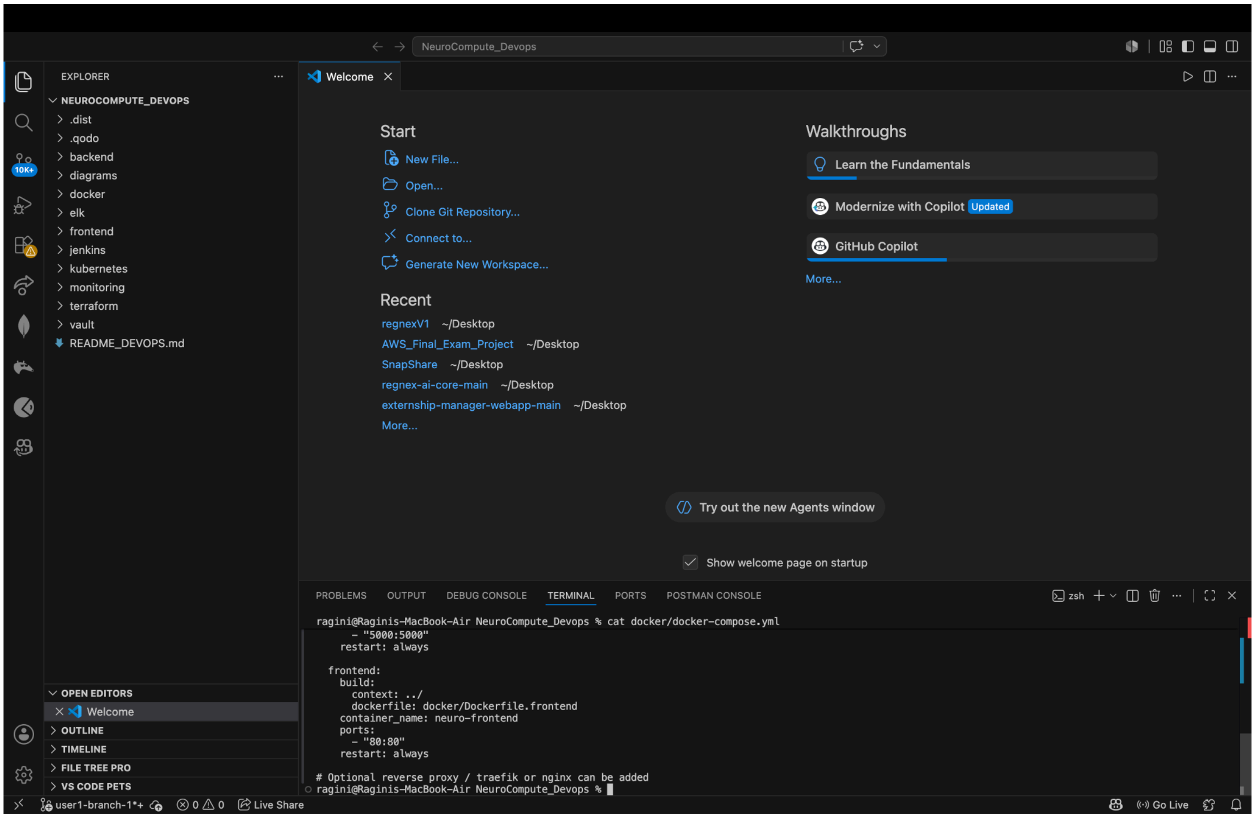
4. Search Job



5. Docker Install



6. All Tool Install



Dashboard

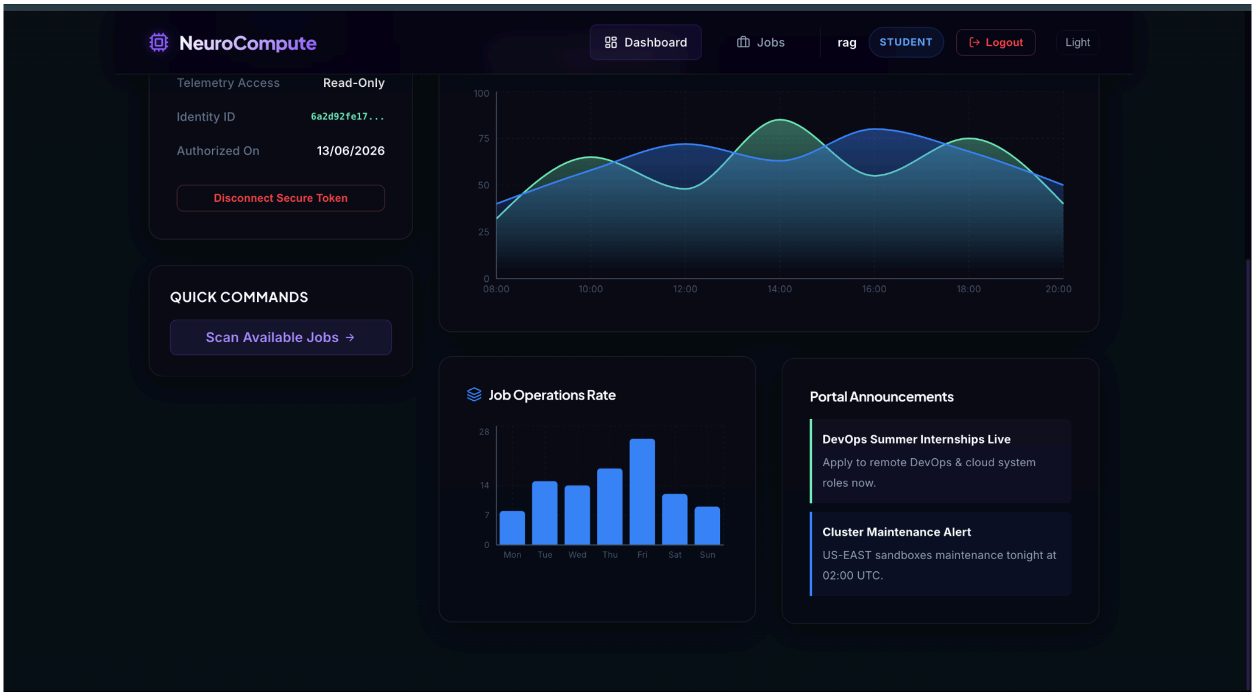
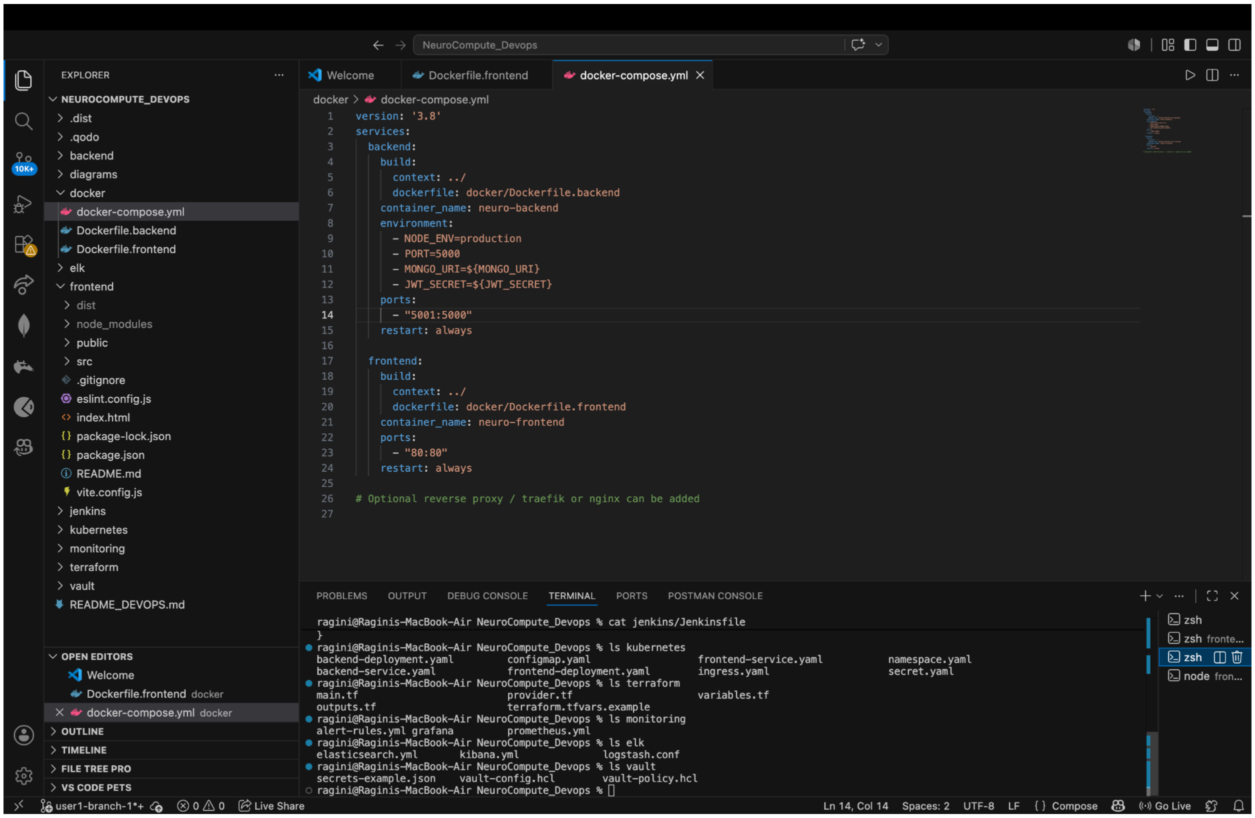
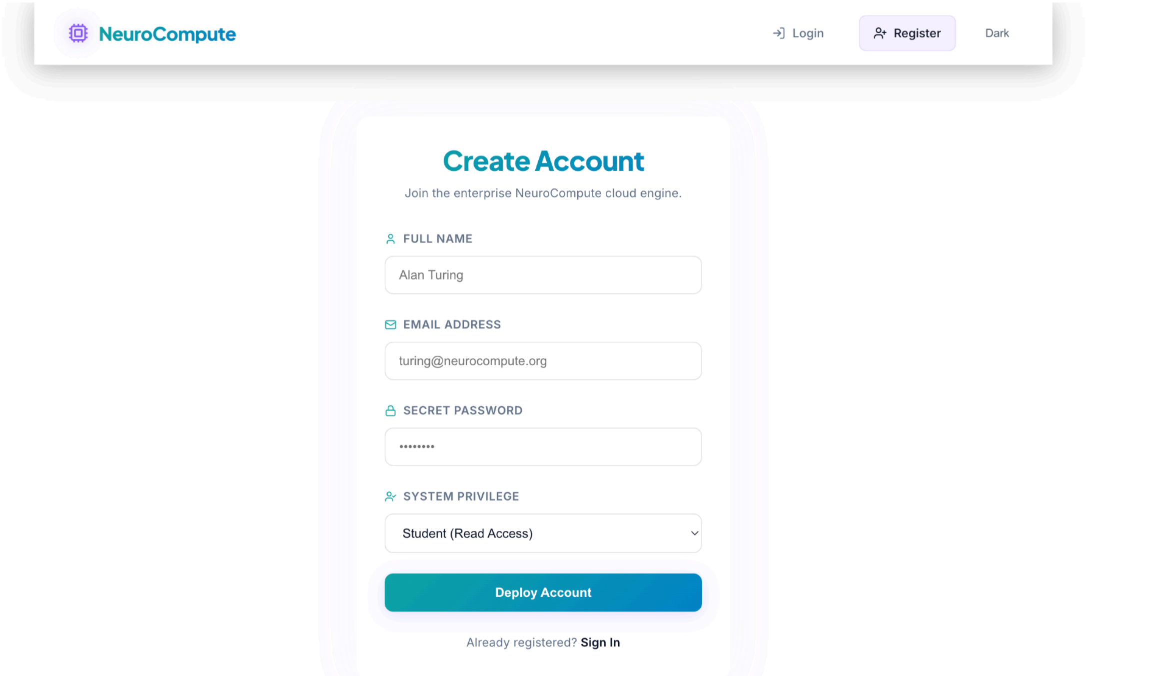

NeuroCompute
Dashboard
Jobs
rag
STUDENT
Logout
Light
Telemetry Access
Read-Only
Identity ID
6a2d92fe17...
Authorized On
13/06/2026
Disconnect Secure Token
QUICK COMMANDS
Scan Available Jobs →
100
75
50
25
0
08:00
10:00
12:00
14:00
16:00
18:00
20:00
Job Operations Rate
28
14
7
0
Mon
Tue
Wed
Thu
Fri
Sat
Sun
Portal Announcements
DevOps Summer Internships Live
Apply to remote DevOps & cloud system roles now.
Cluster Maintenance Alert
US-EAST sandboxes maintenance tonight at 02:00 UTC.

Fig : Docker Configuration , Jenkins Pipeline, Kubernetes Files, Terraform Files, Monitoring Setup, ELK Stack Configuration



Registration Page



Challenges Faced

- Docker image build issues
- Dependency compatibility issues
- Container networking setup
- Kubernetes configuration management
- MongoDB connection troubleshooting
- Environment variable handling

Outcomes

The project successfully demonstrates:

- Full Stack Development
- DevOps Practices
- Cloud Computing Concepts

- Infrastructure Automation
- CI/CD Pipelines
- Container Orchestration
- Monitoring & Logging
- Secret Management

Future Enhancements

- Production Kubernetes Cluster Deployment
- Helm Charts
- GitHub Actions Integration
- AWS Deployment
- Auto Scaling
- Advanced Monitoring
- Disaster Recovery Setup

Conclusion

NeuroCompute DevOps Platform successfully integrates modern cloud-native technologies and DevOps practices into a single project. The implementation demonstrates containerization, CI/CD automation, orchestration, infrastructure provisioning, monitoring, logging, and security management. The project serves as a practical example of industry-standard DevOps workflows and provides a strong foundation for production-grade deployments.

References

1. Docker Documentation
2. Jenkins Documentation
3. Kubernetes Documentation
4. Terraform Documentation
5. Prometheus Documentation
6. Grafana Documentation
7. Elastic Documentation
8. HashiCorp Vault Documentation
9. MongoDB Atlas Documentation

10. React Documentation

THANKYOU!!!!!